

**Vidya Vikas Institute of Engineering and Technology,
Mysore
Library and Information Centre**



VTU Question Papers

MASTER OF COMPUTER APPLICATION

Even Semester

2021-22

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

INDEX

S/L	Subject Code	Subject Title	Exam Dates
1	20MCA11	DATA STRUCTURES WITH ALGORITHMS	July/Aug. 2022
2	20MCA12	OPERATING SYSTEM WITH UNIX	July/Aug. 2022
3	20MCA13	COMPUTER NETWORKS	July/Aug. 2022
4	20MCA14	MATHEMATICAL FOUNDATION FOR COMPUTER APPLICATIONS	July/Aug. 2022
5	20MCA15	RESEARCH METHODOLOGY AND IPR	July/Aug. 2022
6	20MCA21	DATABASE MANAGEMENT SYSTEM	July/Aug. 2022
7	20MCA22	OBJECT ORIENTED PROGRAMMING WITH JAVA	July/Aug. 2022
8	20MCA23	WEB TECHNOLOGIES	July/Aug. 2022
9	20MCA24	SOFTWARE ENGINEERING	July/Aug. 2022
10	20MCA251	CYBER SECURITY	July/Aug. 2022
11	20MCA263	MOBILE APPLICATIONS DEVELOPMENT	July/Aug. 2022
12	20MCA41	ADVANCES IN WEB TECHNOLOGIES	July/Aug. 2022
13	20MCA42	PROGRAMMING USING C#	July/Aug. 2022
14	20MCA51	PROGRAMMING USING C# .NET	July/Aug. 2022
15	20MCA52	MOBILE APPLICATIONS	July/Aug. 2022
16	20MCA53	MACHINE LEARNING	July/Aug. 2022

--	--	--	--	--	--	--	--	--	--

First Semester MCA Degree Examination, July/August 2022

Data Structures with Algorithms

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is data structure? Explain the classification of data structure with example. (06 Marks)
- b. Write the algorithm to convert infix to postfix expression. (06 Marks)
- c. Write a C program to perform push() and pop() operation in a stack using arrays. (08 Marks)

OR

- 2 a. What is polish and reverse polish expression? Write an algorithm to convert infix expression to prefix expression. (07 Marks)
- b. Write a C program to evaluate the postfix expression. (07 Marks)
- c. Convert the following infix expression to postfix expression:
 - i) $(A + B * C) / (D * P - E * F)$
 - ii) $A * B / C * P + D * E * P / G$ (06 Marks)

Module-2

- 3 a. What is recursion? Write a function to implement tower of Hanoi problem with recursion. (06 Marks)
- b. Write a program to implement double ended queue using array. (08 Marks)
- c. What is priority queue? Write an algorithm to insert the element in priority queue based on priority. (06 Marks)

OR

- 4 a. What is queue? Write the applications of queue. Write the algorithm to insert and delete an item from queue. (08 Marks)
- b. Write a program to implement circular queue. (08 Marks)
- c. What are the advantages of circular queue over general queue? (04 Marks)

Module-3

- 5 a. Explain different types of memory management functions with example. (10 Marks)
- b. Write a program to insert and delete a node from beginning and end of the linked list. (10 Marks)

OR

- 6 a. Explain getnode() and freenode() operations. (06 Marks)
- b. Write an algorithm to delete a node from the middle of the linked list. (08 Marks)
- c. Discuss about static and dynamic memory allocation. (06 Marks)

Module-4

- 7 a. Discuss asymptotic notations and basic efficiency classes with example. (10 Marks)
b. Explain the various stages of the algorithm design and analysis process with the help of a flowchart. What is the efficiency of algorithm? (10 Marks)

OR

- 8 a. What is an algorithm? What are the characteristics of a good algorithm? Explain with example. Also explain algorithm correctness. (10 Marks)
b. Explain the methods to analyze recursive and non-recursive algorithms with example. (10 Marks)

Module-5

- 9 a. Write the algorithm for selection sort and derive the time complexity. (08 Marks)
b. Explain Prim's and Kruskal's algorithm, with example. (12 Marks)

OR

- 10 a. Write a function for quick sort. Also analyze its average case time efficiency using recurrence relations. (08 Marks)
b. Compare divide and conquer method and greedy method of solving problems with suitable examples. (08 Marks)
c. Explain how topological sort algorithm is different from other sorting algorithms. (04 Marks)

* * * * *

CBGS SCHEME

USN

--	--	--	--	--	--	--	--	--

20MCA12

First Semester MCA Degree Examination, July/August 2022

Operating System with UNIX

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define operating system. Different single processor and multiprocessor in detail. (10 Marks)
b. What is system call? Explain the different types of system calls. (10 Marks)

OR

- 2 a. Illustrate the different scheduling algorithms with examples. (10 Marks)
b. Explain the three classical problems of synchronization. (10 Marks)

Module-2

- 3 a. What is a deadlock? What are the necessary conditions for deadlocks? (06 Marks)
b. Explain the Banker's algorithm with suitable examples. (10 Marks)
c. Illustrate the resource-allocation graph. (04 Marks)

OR

- 4 a. Explain the concepts of contiguous memory allocation. (10 Marks)
b. For a given reference string calculate the number of page faults using :
i) Optimal Page Replacement ii) Least Recently Used
7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1. (10 Marks)

Module-3

- 5 a. What is a file? Explain absolute and relative path names. (10 Marks)
b. Briefly explain the Unix file system.(10 Marks)

OR

- 6 a. Illustrate the three standard files. (10 Marks)
b. Differentiate between hard links and symbolic links with an example. (10 Marks)

Module-4

- 7 a. What is a process? Explain the mechanism of process creation. (10 Marks)
b. Briefly explain the difference between internal and external commands. (06 Marks)
c. Write the details of killing processes with signals. (04 Marks)

OR

- 8 a. With syntax and example, explain CASE conditional statement. (10 Marks)
b. Illustrate different operations of test command. (10 Marks)

Module-5

- 9 a. What is awk? Explain simple awk filtering with splitting a line into fields. (10 Marks)
b. Describe : i) Begin and End sections ii) Variable and Expressions. (10 Marks)

OR

- 10 a. Write short notes with respect to : i) Arrays ii) Functions. (10 Marks)
b. Write an awk script to compute gross salary of an employee accordingly to a given rule.
HRA – 15%, DA – 45% of basic salary. (10 Marks)

* * * * *

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. $42+8=50$, will be treated as malpractice.

--	--	--	--	--	--	--	--	--	--

First Semester MCA Degree Examination, July/August 2022

Computer Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain OSI network architecture with a neat diagram. (10 Marks)
- b. Describe the network requirements of application programmer, network operator and network designer with an example. (06 Marks)
- c. Briefly explain applications of computer networks. (04 Marks)

OR

- 2 a. Explain following ideas with respect to network architecture :
layering, protocol, encapsulation, multiplexing and demultiplexing. (10 Marks)
- b. Consider a point-to-point link 4 km in length. At what bandwidth would propagation delay (at a speed of 2×10^8 m/s) equal transmit delay for 100-byte packets? (04 Marks)
- c. Explain circuit switched and packet switched networks with a neat diagram. (06 Marks)

Module-2

- 3 a. Apply NRZ, NRZI, Manchester encoding to the following bit stream :
0 0 1 0 1 1 1 1 0 1 0 0 0 0 1 0 (06 Marks)
- b. Briefly explain Byte oriented protocols. (06 Marks)
- c. Discuss briefly about wifi and Bluetooth. (08 Marks)

OR

- 4 a. Explain the concept of stop and wait protocol with necessary diagrams for different scenario. (10 Marks)
- b. A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is $x^3 + 1$.
(i) What is the actual bit string transmitted?
(ii) Suppose the third bit from the left is inverted during transmission. How will receiver detect this error? (10 Marks)

Module-3

- 5 a. Define virtual circuit identifier? Explain virtual circuit network and VC table entry for switches with a neat diagram. (10 Marks)
- b. Define spanning tree algorithm and explain how it works. (10 Marks)

OR

- 6 a. Explain IPV₄ packet headers format with a neat diagram. (10 Marks)
- b. Briefly discuss about,
(i) ARP
(ii) ICMP
(iii) DHCP
(iv) Border Gateway Protocol (10 Marks)

Module-4

- 7 a. Explain three-way hand shakes algorithm for connection establishment and termination respectively for TCP with diagram. (10 Marks)
b. Explain UDP header format. (06 Marks)
c. What are the key characteristics of UDP? (04 Marks)

OR

- 8 a. Explain congestion avoidance mechanisms in brief. (10 Marks)
b. Briefly explain the mechanisms used for TCP congestion control. (10 Marks)

Module-5

- 9 a. Write a brief note on Domain Name System (DNS). (10 Marks)
b. Explain the following cryptographic building blocks :
(i) Symmetric-key ciphers
(ii) Public-key ciphers
(iii) Authenticators
(iv) Ciphers (10 Marks)

OR

- 10 a. What are firewalls? Explain strengths and weaknesses of firewalls. (10 Marks)
b. Explain the following protocols :
(i) SMTP
(ii) HTTP
(iii) SNMP
(iv) IMAP (10 Marks)

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

20MCA14

First Semester MCA Degree Examination, July/August 2022 Mathematical Foundation for Computer Applications

Time: 3 hrs.

Max. Marks:100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Statistical tables are permitted.**

Module-1

- 1 a. Define a set, power set and a singleton set with an example for each. (06 Marks)
- b. State and prove Associative Laws of set theory. (07 Marks)
- c. Find the eigen values and eigen vectors of $A = \begin{pmatrix} -1 & 3 \\ -2 & 4 \end{pmatrix}$. (07 Marks)

OR

- 2 a. Let $A = \{1, 2, 3, 4\}$ $B = \{2, 4, 5, 6\}$. Find : i) $A \cup B$ ii) $A - B$ iii) $B - A$. (07 Marks)
- b. 30 cars were assembled in a factory. The options available were a radio AC and white wall tyres. It is known that 15 of the cars have radios, 8 of them have AC and 6 of them have white wall tyres. Three of them have all 3 options. Determine at least how many cars do not have any option at all. (07 Marks)
- c. State and prove pigeonhole principle. Prove that if 30 dictionaries in a library contain a total of 61, 237 pages then at least one of the dictionaries must have at least 2045 pages. (06 Marks)

Module-2

- 3 a. Define the following with an example :
i) Conjunction
ii) Disjunction. (06 Marks)
- b. Prove that for any 3 propositions p, q, r
 $[p \rightarrow (q \wedge r)] \Leftrightarrow [(p \rightarrow q) \wedge (p \rightarrow r)]$. (07 Marks)
- c. Write the converse, inverse and contrapositive of the statement "If 2 is an integer, then 9 is a multiple of 3". (07 Marks)

OR

- 4 a. Let p, q, r be propositions having the truth values 0, 0 and 1 respectively. Find the truth values of i) $(p \vee q) \vee r$ ii) $(p \wedge q) \wedge r$. (06 Marks)
- b. Give the direct proof and indirect proof of "If n is an odd integer, then n^2 is an odd integer". (05 Marks)
- c. Let the universe be the set of all integers. Consider the following open statements
 $p(x) : x > 3$ $q(x) : x + 1$ is even $r(x) : x \leq 0$.
Write down the truth values of i) $p(2)$ ii) $p(3) \vee \sim r(3)$. (04 Marks)
- d. Test whether the argument is valid. If Sachin hits a century, then he gets a free car.

Sachin hits a century

$\therefore$ Sachin gets a free car

(05 Marks)

Module-3

- 5 a. Let :
 $A = \{1, 2, 3, 4\}$
 $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 3), (3, 3), (4, 4)\}$
 a relation on A. Verify that R is an equivalence relation on A. (06 Marks)
- b. Let $A = \{1, 2, 3, 4\}$ R be the relation on A defined by xRy iff x divides y. Write down R as the set of ordered pairs and draw the dgraph of R. (07 Marks)
- c. Let $M_R = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ $M_S = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ be the matrix relations of the relations R and S on A.
- Find :
 i) $R \cup S$
 ii) $R \cap S$
 iii) $\overline{R}$. Given $A = \{a, b, c\}$. (07 Marks)

OR

- 6 a. Define partition of a set and equivalent class with an example. (10 Marks)
- b. Draw the Hasse diagram representing the positive divisors of 36. (10 Marks)

Module-4

- 7 a. The probability distribution of a finite random variable X is given by the following table :

x	-2	-1	0	1	2	3
P(x)	0.1	K	0.2	2K	0.3	K

- Find the value of K, mean and variance. (10 Marks)
- b. When a coin is tossed 4 times, find the probability of getting.
 i) Exactly one head
 ii) At most 3 heads
 iii) At least 2 heads. (10 Marks)

OR

- 8 a. Find the value of c such that

$$f(x) = \begin{cases} \frac{x}{6} + c & 0 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$
 is a p.d.f. Also find $P(1 \leq x \leq 2)$ (06 Marks)
- b. In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that the shower will last for :
 i) 10 min or more
 ii) Less than 10 minutes
 iii) Between 10 and 12 minutes. (07 Marks)
- c. If x is a normal variable with mean 30 and standard deviation 5 find the probability that
 i) $26 \leq x \leq 40$
 ii) $x \geq 45$. (07 Marks)

Module-5

- 9 a. Define the following with suitable examples :

- i) Simple graph
- ii) Null graph
- iii) Complete bipartite graph.

(06 Marks)

- b. Explain Konigsberg bridge problem.

(07 Marks)

- c. Check whether the following graphs are isomorphic.

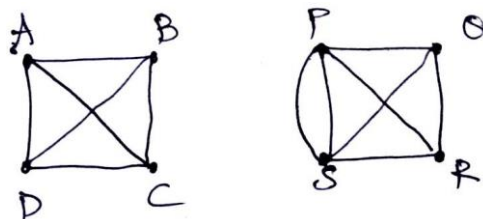


Fig.Q9(c)

(07 Marks)

OR

- 10 a. Define the following with an example :

- i) Spanning subgraph
- ii) Induced subgraph
- iii) Planar graphs.

(06 Marks)

- b. Give the graph coloring of the graph shown in Fig.Q10(b) below.

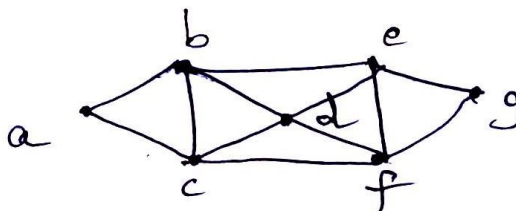


Fig.Q10(b)

(07 Marks)

- c. Use Dijkstra's algorithm to find the shortest path from vertex 1 to each of the other vertices in the weighted directed network shown in the Fig.Q10(c).

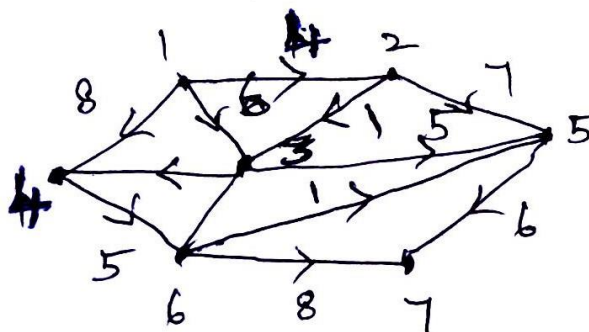


Fig.Q10(c)

Indicate the weight of the shortest paths.

(07 Marks)

USN

--	--	--	--	--	--	--	--	--	--

20MCA15

First Semester MCA Degree Examination, July/August 2022

Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Research? Explain objectives of Research and Motivation in Research. (10 Marks)
- b. Explain the problems encountered by Researchers in India. (10 Marks)

OR

- 2 a. Briefly explain with a neat diagram, Research process with a flow chart. (10 Marks)
- b. Explain the concepts of Research methods Vs Methodology. (10 Marks)

Module-2

- 3 a. What is a Research problem which are the techniques involved in defining problem? Explain. (10 Marks)
- b. Explain the concepts of Reviewing the selected literature. (10 Marks)

OR

- 4 a. How to develop a Theoretical frame work in Reviewing the literature? (10 Marks)
- b. Explain with suitable example, writing about the literature reviewed. (10 Marks)

Module-3

- 5 a. Explain the various concepts relating to Research Design. (10 Marks)
- b. Explain any 2 types of Informal experimental definitions in Research. (10 Marks)

OR

- 6 a. Explain Criteria of selecting a sampling procedure in Research Design. (10 Marks)
- b. Explain with neat diagram of chart showing basic sampling design. (10 Marks)

Module-4

- 7 a. What is a method of Data collection? Explain with a neat diagram of collection of Primary Data. (10 Marks)
- b. Explain selection of Appropriate method in Data collection. (10 Marks)

OR

- 8 a. What is Interpretation? Explain technique of Interpretation. (10 Marks)
- b. Explain different steps in writing Report. (10 Marks)

Module-5

- 9 a. What is Intellectual property? Explain Copyrights and Protectable in Intellectual property. (10 Marks)
- b. Explain International Trade Mark Association and World Intellectual Property Organization. (10 Marks)

OR

- 10 a. What is a Patent? Explain Indian Patent Acts 1970. (10 Marks)
- b. What is a Copy Rights Act? Explain Copy Rights Act 1957. (10 Marks)

* * * * *

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg, 42+8 = 50, will be treated as malpractice.

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

20MCA21

Second Semester MCA Degree Examination, July/August 2022 Database Management System

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Database Management System? Explain Actors on the scene and Workers behind the scene. (10 Marks)
- b. Describe the three – Schema architecture of DBMS with a diagram. (10 Marks)

OR

- 2 a. Explain Database system environment with a diagram. (10 Marks)
- b. Explain centralized and client–server architectures with a diagram. (10 Marks)

Module-2

- 3 a. What is schema? Explain Database schema with an example. (10 Marks)
- b. Explain Primary key and Foreign key with an example. (10 Marks)

OR

- 4 a. What is the Entity? Explain the entity type and Entity sets with a examples. (10 Marks)
- b. What is the need for normalization? Explain 1NF, 2NF with an example. (10 Marks)

Module-3

- 5 a. What is Datatype? Explain different datatypes in SQL. (10 Marks)
- b. Explain constraints in SQL with an example. (10 Marks)

OR

- 6 a. Explain aggregate functions in SQL with an example. (10 Marks)
- b. Explain Group by and having clauses with an example. (10 Marks)

Module-4

- 7 a. What is Trigger? Explain with example. (10 Marks)
- b. Define views in SQL. Explain with example. (10 Marks)

OR

- 8 a. Describe the stored procedure with an example. (10 Marks)
- b. Explain functions with an example. (10 Marks)

Module-5

- 9 a. Define Transaction. Explain ACID properties with examples. (10 Marks)
- b. Explain two–phase locking techniques with an example. (10 Marks)

OR

- 10 a. Explain concurrency control based on timestamp ordering with example. (10 Marks)
- b. What is recovery? Explain recovery in multidatabase systems. (10 Marks)

* * * * *

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg. 42+8 = 50, will be treated as malpractice.

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

20MCA22

Second Semester MCA Degree Examination, July/August 2022 Object Oriented Programming with Java

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the Buzzwords of Java. (10 Marks)
- b. Explain the various primitive data types used in Java. Give suitable examples. (10 Marks)

OR

- 2 a. Define an Array. Write the array declaration syntax for multidimensional arrays. Write a simple Java program to create an array of objects. (10 Marks)
- b. Demonstrate the concept of garbage collection through suitable examples. (10 Marks)

Module-2

- 3 a. Explain method overloading and constructor overloading with suitable examples. (10 Marks)
- b. Explain the following :
 - i) Varargs
 - ii) Final keyword. (10 Marks)

OR

- 4 a. What are super class constructors? How does super class members are used in Java. (10 Marks)
- b. What is Method-Overriding? Explain how it allows Java to support run-time polymorphism with an example. (10 Marks)

Module-3

- 5 a. What are interfaces? What are their benefits? Explain how it is implemented in java with suitable example. (10 Marks)
- b. What is Exception? Explain how exception handling mechanism can be used for debugging a program. (10 Marks)

OR

- 6 a. Define package. Explain the creation of a package using a suitable example program. (10 Marks)
- b. Develop a simple program and explain multiple catch blocks. (10 Marks)

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg. 42+8 = 50, will be treated as malpractice.

Module-4

- 7 a. Define synchronization? Explain how inter-thread communication can be achieved in multithreading using producer and consumer problem. (10 Marks)
b. Define multi-threading? Construct a java program to create multiple threads in Java by implementing runnable interface. (10 Marks)

OR

- 8 a. Develop a simple program and explain how enumerations are class types. (10 Marks)
b. Analyze the following with syntax and example :
i) values() and valuesof()
ii) compareTo(). (10 Marks)

Module-5

- 9 a. What are URL classes? Explain URL connections with syntax. (10 Marks)
b. What HttpURLConnection class. Explain the methods of its. (10 Marks)

OR

- 10 a. Identify different types of methods defined by collection interface. (10 Marks)
b. Demonstrate a simple program, the constructors and methods in ArrayList class. (10 Marks)

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

20MCA23

Second Semester MCA Degree Examination, July/August 2022

WEB Technologies

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Write short notes on the following :
 - i) Web browser
 - ii) Web server
 - iii) MIME

(10 Marks)
- b. Discuss the basic structure of a XHTML document. (05 Marks)
- c. What are the rules to be followed to make use of the html elements in a XHTML document? Explain. (05 Marks)

OR

- 2 a. Demonstrate the use of image maps with a suitable example. (10 Marks)
- b. What is a hyperlink? Explain the various ways of creating hyper links with example. (10 Marks)

Module-2

- 3 a. Mention any five CSS selectors and explain their use with a suitable example. (10 Marks)
- b. Discuss on the different ways of including CSS style information to a HTML document. (10 Marks)

OR

- 4 a. What is an Array in Javascript? Explain the various ways of creating arrays. Mention any 5 array methods and explain their use. (10 Marks)
- b. Write a Javascript program that accepts a value 'n' as input from the user and display the first 'n' fibonacci numbers as output. (05 Marks)
- c. Write a Javascript program that accepts valid USN as input and displays an error message otherwise. USN format 'XX00 XXX 00' where 'X' is an upper case alphabet, '0' is a digit. (05 Marks)

Module-3

- 5 a. What is Bootstrap? What are the features of Bootstrap? Create a simple web page using Bootstrap. (10 Marks)
- b. Discuss on the following :
 - i) Table
 - ii) Image
 - iii) Button
 - iv) Progress bar.

Use Bootstrap code snippet. (10 Marks)

OR

- 6 a. Discuss briefly on the Grid system of Bootstrap. (10 Marks)
b. List out the various types of forms in Bootstrap. Explain their use with code snippet. (10 Marks)

Module-4

- 7 a. What is JQuery? What are the advantages of JQuery? (06 Marks)
b. Explain the syntax of JQuery script with a suitable example. (06 Marks)
c. Write a JQuery program to animate a rectangular Box. (08 Marks)

OR

- 8 a. Mention any ten JQuery selectors and explain their use with code snippet. (10 Marks)
b. Demonstrate the use of any five JQuery method to manipulate HTML elements with code snippet. (05 Marks)
c. Write a JQuery program that performs fading in and fading out of the HTML element. (05 Marks)

Module-5

- 9 a. What is an angular JS Directive? Explain the use of the following directives with code snippet. (08 Marks)
b. What is the use of a filter in Angular JS? Explain the use of JSON, Currency, Number, Lowercase. (08 Marks)
c. Demonstrate \$ scope with an example. (04 Marks)

OR

- 10 a. Write an Angular JS program to demonstrate client-side form validation. (10 Marks)
b. With code snippet, explain the use of controllers in Angular JS. (10 Marks)

* * * * *

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

20MCA24

Second Semester MCA Degree Examination, July/August 2022 Software Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Software Engineering. Elaborate upon the essential attributes of a good software. (10 Marks)
b. Illustrate how software engineering concept is applied using the case study of an insulin pump control system. (10 Marks)

OR

- 2 a. Describe the waterfall model with a neat diagram with its merits and demerits. (10 Marks)
b. Explain the principles of agile methods and the extreme programming practices. (10 Marks)

Module-2

- 3 a. What is requirement engineering? Explain the functional and non-functional requirements. (10 Marks)
b. Explain requirements discovery in detail. (10 Marks)

OR

- 4 a. What is SRS document? Explain the structure of SRS document. Also discuss about the users of a requirements document. (10 Marks)
b. Explain requirements validation and also the techniques used for requirements validation. (10 Marks)

Module-3

- 5 a. What is object orientation? Explain the themes of object oriented methodology. (10 Marks)
b. Describe the steps to be followed in Object Oriented development. (10 Marks)

OR

- 6 a. What are links and associations? Write and explain UML notations for links and association with example. (10 Marks)
b. Discuss generalizations, specialization and inheritance concepts. (10 Marks)

Module-4

- 7 a. What are events and states? With necessary examples explain the different types of events. (10 Marks)
b. Draw a nested state diagram for telephone line system. Write transitions with reference to the state diagram behavior. (10 Marks)

OR

- 8 a. What is use case model? Write and explain the use case diagram for a vending machine. Write the guidelines for use case model. (10 Marks)
b. Explain the sequence model for the online stock broker system with a neat diagram. (10 Marks)

Module-5

- 9 a. Describe the two commonly used approaches for effort estimation. (10 Marks)
b. Discuss in detail about the concept of Risk Management. (10 Marks)

OR

- 10** a. Explain the Project Scheduling Process. (10 Marks)
b. Explain Configuration Management Planning. (10 Marks)

* * * * *

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

20MCA251

Second Semester MCA Degree Examination, July/August 2022 Cyber Security

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Cybercrime. List and explain how Cybercrime are classified. (10 Marks)
- b. Explain the different categories of Cybercrimes. (05 Marks)
- c. Explain the different phases involved in Cybercrime. (05 Marks)

OR

- 2 a. What are attack vector? How are they launched? (10 Marks)
- b. Define Botnet. Explain the different ways to ensure that your system is secure. (10 Marks)

Module-2

- 3 a. Define Password cracking. Explain the different password guidelines to avoid account hacked by the attacker. (10 Marks)
- b. Define Viruses. List and explain the different types of viruses. (10 Marks)

OR

- 4 a. What do you mean by DOS attack? Explain different types or levels of DOS attacks. (10 Marks)
- b. Define Buffer Overflow. List and explain different types of Buffer Overflow. Add a note on minimize the Buffer Overflow. (10 Marks)

Module-3

- 5 a. What is Phishing? Explain the different types of Phishing seams. (10 Marks)
- b. Define Identity Theft. Explain the different types of Identity Theft. (10 Marks)

OR

- 6 a. List and explain the different phases of Digital forensic life cycle. (10 Marks)
- b. What are the guidelines for the (digital) evidence collection phase? (10 Marks)

Module-4

- 7 a. Define Vishing. Explain how it works and how to protect from Vishing attacks. (10 Marks)
- b. With a neat diagram, show the flow of events with CLEW (Closed Loop Environment for Wireless). (05 Marks)
- c. What is Proliferation of mobile and ireless devices? Explain. (05 Marks)

OR

- 8 a. Explain about the Credit Card fraud in mobile and wireless computing era. (10 Marks)
- b. List and explain the popular types of attacks against 3G mobile networks. (05 Marks)
- c. Write a note on Bluetooth Hacking. (05 Marks)

Module-5

- 9 a. Define Firewall. Explain how Firewall protects a Network. (10 Marks)
- b. Write a note on Linux System Firewall and OS X System Firewall. (10 Marks)

OR

- 10 a. Explain VPN and Snort rules in detail. (10 Marks)
- b. Explain Packet filters and NAT. (05 Marks)
- c. Write a note on Block Chain technology. (05 Marks)

* * * * *

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg. 42+8 = 50, will be treated as malpractice.

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

20MCA263

Second Semester MCA Degree Examination, July/August 2022 Mobile Applications Development

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What are the preliminary considerations for Mobile Application Development? (10 Marks)
- b. Demonstrate effective use of Screen Rent Estate. (06 Marks)
- c. List out any 4 features of Android OS, with respect to user interface design. (04 Marks)

OR

- 2 a. Discuss briefly on the design patterns for Navigation used in mobile user interface design. (10 Marks)
- b. Explain Gestalt Principles (06 Marks)
- c. Compare Simulators and Emulators. (04 Marks)

Module-2

- 3 a. Explain the basic components of an Android Application (06 Marks)
- b. Write short notes on the anatomy of an Android Application. (10 Marks)
- c. What is a fragment? How is it useful in designing an application? (04 Marks)

OR

- 4 a. What is an action bar? Demonstrate how to utilize it with an example. (08 Marks)
- b. What is Android Manifest File? Explain its contents in detail. (08 Marks)
- c. List out the tools required for developing an Androids application. (04 Marks)

Module-3

- 5 a. Illustrate the use of a list view with a code snippet. (10 Marks)
- b. With an example application demonstrate the use of Checkbox, Radiobutton, Button EditText and TextVeiw. (10 Marks)

OR

- 6 a. Explain the steps involved in displaying maps in an android application. (10 Marks)
- b. Discuss on the steps involved in publishing an android application in Google Playstore. (10 Marks)

Module-4

- 7 a. Define service? Explain how an activity can invoke a service with an example. (10 Marks)
- b. Give the code snippet for downloading an image from a given URL and display it using ImageView. (10 Marks)

OR

- 8 a. Explain briefly how SMS works in Android. (10 Marks)
- b. Write short notes on Asynchronous tasks in Android. (08 Marks)
- c. Name the permissions you need to declare in your Android Manifest file for HTTP connection. (02 Marks)

Module-5

- 9** a. Explain briefly the steps to create native iOS applications. (10 Marks)
b. Discuss on the anatomy of an iOS application. (10 Marks)

OR

- 10** a. Explain the various types of Notifications used in Windows Phone 7. (08 Marks)
b. Write short note on the following with respect to windows Phone 7.
i) Offline Storage ii) Accelerometer. (08 Marks)
c. Explain GPS features for Windows Phone 7 Application. (04 Marks)

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

20MCA41

Fourth Semester MCA Degree Examination, July/August 2022

Advances in Web Technologies

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain various scalar types and operators available in PHP. (10 Marks)
- b. Explain the various looping statements in PHP. Write any PHP code to explain 'for' loop. (10 Marks)

OR

- 2 a. Outline file handling in PHP. (10 Marks)
- b. Illustrate steps to join web form with database in PHP. (06 Marks)
- c. Write a short note on Cookies in PHP. (04 Marks)

Module-2

- 3 a. Illustrate with syntax any eight built-in-methods for arrays in Ruby. (08 Marks)
- b. Explain codeblocks and iterators with sample code in Ruby. (06 Marks)
- c. Summarize the following string methods i) Capitalize ii) Strip iii) Chop. (06 Marks)

OR

- 4 a. With a neat diagram, explain directory structure for Rails application. (10 Marks)
- b. Illustrate the concept of classes in Ruby with an example. (10 Marks)

Module-3

- 5 a. What is Ajax? Explain with an diagram, how it is different from traditional web applications. (10 Marks)
- b. With a neat diagram, explain Ajax web application model. (10 Marks)

OR

- 6 a. With an example illustrate sending data to server using methods of GET and POST for XMLHttpRequest object. (10 Marks)
- b. List and explain Ajax principles. (06 Marks)
- c. Differentiate between synchronous and Asynchronous requests. (04 Marks)

Module-4

- 7 a. Explain how to handle multiple XMLHttpRequest objects in the same page. (08 Marks)
- b. Explain the pattern of predictive fetch using Ajax. (08 Marks)
- c. Write a note on submission throttling. (04 Marks)

OR

- 8 a. With example, explain pattern of periodic refresh and fall back pattern in Ajax. (10 Marks)
- b. Illustrate how to implement the Ajax pattern of multistage download. (10 Marks)

Module-5

- 9 a. What is Bootstrap? Explain the Bootstrap life structure with a neat diagram and give an example of basic HTML template using Bootstrap. (10 Marks)
- b. With a sample code explain Fluid Grid system and container layouts. (10 Marks)

OR

- 10 a. Explain with an example, the Bootstrap Button classes for different styles of Button in Bootstrap CSS. (10 Marks)
- b. Explain the following HTML concepts with Bootstrap CSS, i) Headings ii) Emphasis classes. (10 Marks)

* * * * *

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. 42+8 = 50, will be treated as malpractice.

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

20MCA42

Fourth Semester MCA Degree Examination, July/August 2022 Programming using C#

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the components of .Net framework with the help of architecture diagram. (10 Marks)
- b. List and explain the different data types in C#. (10 Marks)

OR

- 2 a. Discuss between value types of reference types and write a program to explain boxing and unboxing. (10 Marks)
- b. Write short notes on :
 - i) Windows work flow foundation
 - ii) Windows cold space and LINQ. (10 Marks)

Module-2

- 3 a. Explain partial classes and partial methods, with the help of a C# program. (10 Marks)
- b. Write a C# program to explain accessor and mutator properties used in encapsulation. (10 Marks)

OR

- 4 a. Explain the significance of “is – a” and “has – a” relationship in inheritance with an example program. (10 Marks)
- b. Explain the following with example :
 - i) Static class and static members
 - ii) Abstract class and abstract methods. (10 Marks)

Module-3

- 5 a. Explain the four steps involved in creating and using delegate in the program. (10 Marks)
- b. Explain the architecture of ADO.NET with a neat diagram and list the ADO.NET provided. (10 Marks)

OR

- 6 a. Write a C# program using try, catch, finally to explain any predefined exceptions. (10 Marks)
- b. Explain the properties and methods of Data Reader and Data Adaptor class. (10 Marks)

Module-4

- 7 a. Explain various mouse events in C# windows application. (10 Marks)
- b. Explain the WPF architecture with a neat diagram. (10 Marks)

OR

- 8 a. Explain the following :
i) XAML elements
ii) Makeup extension classes in XAML. (10 Marks)
- b. Write short notes on :
i) MDI window forms
ii) Event –Driven GUI. (10 Marks)

Module-5

- 9 a. Explain different validation controls with suitable example supported by ASP.NET. (10 Marks)
- b. Explain the architecture of three tier web based application. (10 Marks)

OR

- 10 a. What are Cookies? Explain session management in ASP.NET using cookies. (10 Marks)
- b. Explain the controls from AJAX control toolkit. (10 Marks)

* * * * *

--	--	--	--	--	--	--	--	--	--

Fifth Semester MCA Degree Examination, July/August 2022

Programming using C# .Net

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a neat diagram, explain the .NET execution engine. (10 Marks)
- b. Write a short note on Windows Workflow Foundation. (05 Marks)
- c. Explain the benefits of .NET Framework. (05 Marks)

OR

- 2 a. Develop a C# program to boxing, unboxing and bad-boxing. List out the difference between value type and reference type. (10 Marks)
- b. What is an Assembly? Describe the information stored in an assembly. (06 Marks)
- c. Write a short note on namespace. (04 Marks)

Module-2

- 3 a. Discuss the different ways of enforcing the encapsulation with examples for each. (10 Marks)
- b. Along with an example, explain “Partial” keyword importance. (05 Marks)
- c. Demonstrate the indexes with an example. (05 Marks)

OR

- 4 a. Discuss the concept of sealed class and sealed method suitable example. (08 Marks)
- b. List and explain the different types of access modifiers. (06 Marks)
- c. Develop a C# program to overload the + operators. (06 Marks)

Module-3

- 5 a. Describe the four steps to create and use a delegate in your program with an example. (10 Marks)
- b. Explain how custom exceptions will be created in C# with suitable example. (10 Marks)

OR

- 6 a. Describe the architecture of ADO.NET with a neat diagram in detail. (10 Marks)
- b. Explain the procedure of getting connected to a database and running the following queries with relevant example:
 - (i) Insert record to table.
 - (ii) Delete records from a table. (10 Marks)

Module-4

- 7 a. What is GUI? List and explain basic GUI controls of Window forms. (10 Marks)
- b. Write a C# program for addition of two numbers from text boxes display the result in form using button click event. (06 Marks)
- c. Write a short note on MDI window. (04 Marks)

OR

- | | | |
|----------|--|------------|
| 8 | a. Discuss the WPF Architecture with a neat diagram. | (10 Marks) |
| | b. Explain XAML elements with an example. | (05 Marks) |
| | c. Explain markup extension classes in XAML. | (05 Marks) |

Module-5

- | | | |
|----------|--|------------|
| 9 | a. Discuss the Multi-Tier application architecture of ASP.Net with a neat diagram. | (10 Marks) |
| | b. Explain the session management in ASP.Net using controls. | (10 Marks) |

OR

- | | | |
|-----------|---|------------|
| 10 | a. Describe the different validation controls in ASP.NET. | (10 Marks) |
| | b. Illustrate the controls from AJAX control toolkit. | (10 Marks) |

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

18MCA52

Fifth Semester MCA Degree Examination, July/August 2022

Mobile Applications

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Mobile Application? (02 Marks)
 b. Explain in detail about the hardware and software influenced for cost of development of a Mobile Application development. (08 Marks)
 c. Write various market competitors to Android and explain about any two third – party frameworks of Mobile Application development. (10 Marks)

OR

- 2 a. Discuss any five main type of Mobile Application in detail with an example each. (10 Marks)
 b. Explain in detail of your understanding of Mobile Applications users according to Gestalt principles with an example each. (10 Marks)

Module-2

- 3 a. Draw an Android Architecture diagram and explain each layer in detail. (10 Marks)
 b. Describe Anatomy of Android Project structure. (06 Marks)
 c. Illustrate and write the tools and software used to develop an Android application. (04 Marks)

OR

- 4 a. With a neat diagram, explain each phases of activity Life Cycle with a Snippet code. (12 Marks)
 b. Explain View and View Groups in Android Application, with a proper syntax and examples. (08 Marks)

Module-3

- 5 a. Define Fragment. Write the types of Fragments in Android. (04 Marks)
 b. Illustrate the List Fragment with a Snippet Code. (06 Marks)
 c. Discuss in detail, all 5 types of Layout in Android. (10 Marks)

OR

- 6 a. Write a short note on the following :
 i) Toggle Button view i) Date Picker view
 iii) Auto complete Text view iv) Progress Bar view. (12 Marks)
 b. Discuss in detail, how Google Map is used in the Android activity. (08 Marks)

Module-4

- 7 a. Explain in detail, how SMS and E – mail works in Android Application. (12 Marks)
 b. Write a note on the following (any two) :
 i) Relative view ii) Scroll view
 iii) Intent iv) Toast. (08 Marks)

OR

- 8 a. Write the code to handle the user preferences selected by user and display the selected preferences either by using alert or toast notification. (08 Marks)
b. Write a code for Android application to download an image file from the Web. (08 Marks)
c. Briefly discuss the concept of binding activities to Services. (04 Marks)

Module-5

- 9 a. Draw a neat diagram of Windows Phone 7 storyboard ; where you will have a better conceptual overview of all the views and connection between them in the VTI Exam application form. (10 Marks)
b. How Windows Phone is distinct from Android? Justify your answer with atleast by discussing 3 feature of Windows Phone. (08 Marks)
c. What do you understand about the Project structure of Windows Phone? Write in brief. (02 Marks)

OR

- 10 a. Write the differences between iOS and Windows Phone 7. Justify your answer with 3 features of iOS Phone. (08 Marks)
b. Write Anatomy structure of an iOS App fro iOS Project. (08 Marks)
c. Write the usage of any two components of iOS Project. (04 Marks)

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

18MCA53

Fifth Semester MCA Degree Examination, July/August 2022 Machine Learning

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define machine learning and explain well-posed learning problem with example. (05 Marks)
- b. Explain various stages involved in designing learning system. (10 Marks)
- c. Write Finds Algorithm and explain. (05 Marks)

OR

- 2 a. Define concept learning and discuss inductive learning hypothesis. (04 Marks)
- b. Give candidate elimination algorithm and find maximally general and specific hypothesis for the given training data set.

Sky	Temp	Humidity	Wind	Water	Forecast	Enjoy
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	Strong	Strong	Warm	Charge	No
Sunny	Warm	High	Strong	Cool	Charge	Yes

- c. Discuss various issues in machine learning. (10 Marks)

(06 Marks)

Module-2

- 3 a. State Entropy and Information Gain. (02 Marks)
- b. Construct Decision Tree for the following Boolean functions :
i) $A \wedge \neg B$ ii) $A \vee (B \wedge C)$ iii) $A \text{ XOR } B$. (06 Marks)
- c. Discuss issues of avoiding over fitting of data, handling continuous attribute and missing attribute values in decision trees. (12 Marks)

OR

- 4 a. Discuss inductive Bias in Decision Tree learning and differentiate between two types of Bias. (04 Marks)
- b. Describe ID₃ algorithm for decision tree learning. (08 Marks)
- c. Explain various appropriate problems for decision tree learning. (08 Marks)

Module-3

- 5 a. What are the types of problems in which ANN can be applied? (08 Marks)
- b. Give the algorithm for Back propagation. (06 Marks)
- c. Explain the concept of perceptron with neat diagram. (06 Marks)

OR

- 6 a. What is Gradient Descent? Write gradient descent algorithm for training a linear unit. (08 Marks)
- b. Discuss stochastic approximation to Gradient Descent. (08 Marks)
- c. Differentiate between perceptron training rule and delta rule. (04 Marks)

Module-4

- 7 a. Define Bayes theorem. What is relevance and features of Bayesian theorem? (08 Marks)
 b. Explain the concept of EM Algorithm. (04 Marks)
 c. Explain Brute force Bayes concept learning with algorithm. (08 Marks)

OR

- 8 a. Consider the given dataset. Apply Naïve Baye's algorithm and predict that if a fruit has the following properties then which type of fruit it is. Fruit = {yellow, sweet, long}.
 Given :

Fruit	Yellow	Sweet	Long	Total
Mango	350	450	0	650
Banana	400	300	350	150
Others	50	100	50	1200

- b. Explain the concept of minimum description length principle. (10 Marks)
 c. Explain Bayesian belief nets and conditional independence with example. (05 Marks)
 (05 Marks)

Module-5

- 9 a. Define Simple error and True error. (04 Marks)
 b. Explain K – nearest neighbor learning algorithm. (08 Marks)
 c. What are instance based learning? Explain key features and disadvantage of these methods. (08 Marks)

OR

- 10 a. Define variance, standard deviation and estimate bias of random variable. (04 Marks)
 b. What is reinforcement learning? How it differs from other approximation tasks. (08 Marks)
 c. Explain the Q-function Q-learning algorithm assuming deterministic rewards and actions. (08 Marks)

* * * * *